

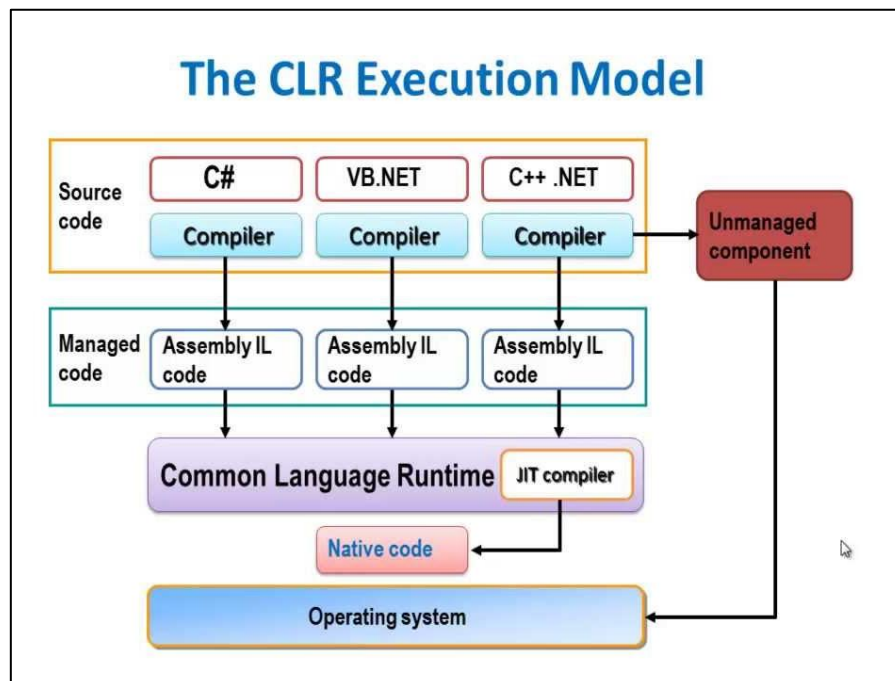
.NET & C# Fundamentals

1. .NET Architecture

What is .NET?

- .NET is an **open-source, cross-platform development platform** developed by Microsoft.
- It is used to build:
 - Console Applications
 - Desktop Applications
 - Web Applications
 - Web APIs
 - Microservices (Mini-APIs)
 - Mobile Apps
 - Cloud Applications

.NET Execution Flow



C# Source Code



C# Compiler (Roslyn)



Intermediate Language (IL/MSIL)



CLR (JIT Compiler)



Native Machine Code



Operating System

Components of .NET

- Programming Language (C#, F#, VB.NET)
- Compiler
- Intermediate Language (IL)
- Common Language Runtime (CLR)
- Base Class Library (BCL)

Advantages

- Cross-platform
- Object-Oriented
- Secure
- Automatic Memory Management
- High Performance
- Rich Class Library

2. CLR (Common Language Runtime)

Definition

CLR is the **runtime environment** that executes .NET applications.

Think of CLR as the **engine** of the .NET platform.

Responsibilities of CLR

- Executes .NET programs
- JIT Compilation
- Memory Management
- Garbage Collection
- Exception Handling
- Security
- Thread Management
- Type Safety
- Debugging Support

JIT Compiler

- JIT stands for **Just-In-Time Compiler**.
- Converts IL code into native machine code at runtime.
- Only required methods are compiled.

Garbage Collector (GC)

- Automatically removes unused objects from memory.
- Prevents memory leaks.
- Developers do not need to manually free memory.

3. CTS (Common Type System)

Definition

CTS defines **all data types and programming rules** that every .NET language follows.

Purpose

- Ensures language interoperability.
- Provides common data types.

Categories

Value Types

Stored directly in memory (Stack).

Examples: int, double, bool, char, struct, enum

Reference Types

Store references to objects (Heap).

Examples: class, string, array, interface, delegate

Value vs Reference Types

Value Type	Reference Type
Stored in Stack	Stored in Heap
Stores actual value	Stores object reference
Examples: int, bool	Examples: class, string

4. CLS (Common Language Specification)

Definition

CLS is a **subset of CTS**.

It defines a common set of rules that all .NET languages should follow.

Purpose

- Language interoperability
- Code reusability
- Cross-language compatibility

5. Visual Studio IDE Setup

What is Visual Studio?

Visual Studio is Microsoft's **Integrated Development Environment (IDE)** used to develop .NET applications.

Required Workloads

- .NET Desktop Development
- ASP.NET and Web Development

- Azure Development (Optional)

Important Windows

- Solution Explorer
- Toolbox
- Output Window
- Error List
- Properties Window
- Terminal

Useful Shortcuts

Shortcut	Purpose
F5	Run with Debugging
Ctrl + F5	Run without Debugging
Ctrl + Shift + B	Build Project
F9	Toggle Breakpoint
F10	Step Over
F11	Step Into

6. C# Console Project Structure

Typical Project Structure

MyProject

```

|
├── Program.cs
├── MyProject.csproj
├── bin
└── obj

```

Program.cs

Contains the application's entry point.

```
Console.WriteLine("Hello World");
```

.csproj File

Stores project configuration such as:

- Target Framework
- Package References
- Build Settings

bin Folder

Contains compiled executable files.

obj Folder

Contains temporary build files.

7. C# Programming Fundamentals

Variables

Used to store data.

```
int age = 25;
```

```
string name = "John";
```

Common Data Types

Type	Example
int	100
double	12.5
decimal	5000.75M
bool	true
char	'A'
string	"Hello"

Operators

Arithmetic

+

-

*

/

%

Comparison

==

!=

>

<

>=

<=

Logical

&&

||

!

Assignment

=

+=

-=

*=

Input

```
string name = Console.ReadLine();
```

Output

```
Console.WriteLine(name);
```

String Interpolation

```
Console.WriteLine($"Welcome {name}");
```

Constants

```
const double PI = 3.14;
```

Type Casting

Implicit

```
int x = 10;
```

```
double y = x;
```

Explicit

```
double d = 10.5;
```

```
int x = (int)d;
```

Parsing User Input

```
int age = int.Parse(Console.ReadLine());
```

Safer Option

```
int.TryParse(Console.ReadLine(), out int age);
```

var Keyword

```
var city = "Hyderabad";
```

The compiler automatically determines the variable type.

Comments

```
// Single-line comment
```

```
/* Multi-line comment */
```

Summary for Quick Revision

- **.NET** → Development platform.
- **CLR** → Executes .NET applications.
- **IL (MSIL/CIL)** → Intermediate code generated by the compiler.
- **JIT** → Converts IL to native machine code at runtime.
- **GC** → Automatically frees unused memory.
- **CTS** → Defines all .NET data types.
- **CLS** → Common rules for all .NET languages.
- **BCL** → Built-in reusable class library.
- **Visual Studio** → IDE for developing .NET applications.
- **Program.cs** → Main entry point of a Console application.
- **.csproj** → Project configuration file.
- **Value Types** → Store actual values.
- **Reference Types** → Store references to objects.
- **var** → Compiler infers the type at compile time.
- **TryParse()** → Preferred method for validating user input.